2 **(a)** **(i)** force proportional / equal to rate of change of momentum (Newton II) M1

force on (block) A equal and opposite to force on (block) B (Newton III) M1
(Do not award this mark if the general definition of third law is given.)

equal magnitude of forces and collision time for A and B hence change in momentum same magnitude A1

opposite forces hence change in momentum opposite A1
(This mark is awarded only if "opposite forces" is explicitly linked to direction of change in momentum.)

(ii) if both blocks at rest, magnitude of change in momentum of A = (–)1.2M AND change in momentum of B = 0.25M M1

not equal hence not possible A1

(b) **(i)** *either* $3M \times 0.40 + M \times (-0.25) = 3M \times 0.20 + Mv$ (conservation of momentum)
or $3M \times (0.20 - 0.40) = -M \times (v + 0.25)$ M1

$v = 0.35 \text{ m s}^{-1}$ A1

(ii) *either* show total kinetic energy after < total kinetic energy before collision

or show relative speed of approach ≠ relative speed of separation M1

inelastic A1

3 (a) (i) point which weight of ship M1